Brooklyn Lane

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EDUCATION

University of Washington — *Ph.D. in Computer Science (Computer Vision & Graphics)* 2002 - 2006

Carnegie Mellon University — *B.S. in Computer Science with Honors* 1998 - 2002

EXPERIENCE

Magic Leap — *Principal AR/VR Architect* 2021 - Present

- Architected spatial computing platform serving 2.3M+ enterprise users across healthcare, manufacturing, and design industries; improved depth perception accuracy by 34% through novel optical calibration algorithms.
- Led cross-functional team of 18 engineers to develop next-generation hand tracking system, reducing latency from 45ms to 12ms and achieving 98.7% gesture recognition accuracy in real-world environments.
- Published 4 peer-reviewed papers at SIGGRAPH and IEEE VR on real-time photogrammetry optimization and persistent scene understanding, garnering 2,100+ citations and establishing industry standards for XR content delivery.

Niantic (Pokémon GO, Lightship) — *Distinguished Engineer - AR Systems* 2018 - 2021

- Designed and implemented Lightship ARDK core rendering engine used by 4,000+ developers, processing 150M+ daily AR sessions and generating $45M in developer ecosystem revenue.
- Pioneered visual-inertial odometry improvements that reduced drift by 62% in GPS-denied environments, enabling accurate waypoint navigation for outdoor AR experiences across 195 countries.
- Mentored 12 staff-level engineers and managed $8.2M R&D budget for spatial awareness research; sponsored 8 successful promotions to senior engineer level.

Qualcomm Innovation Center — *Principal Research Scientist - Immersive Computing* 2014 - 2018

- Invented patented real-time ray-tracing optimization for mobile AR, reducing GPU power consumption by 48% while maintaining visual fidelity; licensed to 12+ semiconductor manufacturers generating $12M annual licensing revenue.
- Collaborated with Khronos Group to author OpenXR compliance specifications adopted by 89% of commercial AR/VR platforms; presented keynote at GDC 2017 reaching 3,200+ attendees.
- Developed computer vision pipeline for robust marker-less tracking achieving 99.2% stability across varying lighting conditions, accelerating adoption in enterprise field service platforms by 3 major customers.

Microsoft Mixed Reality Capture Studios (MRCS) — *Principal Software Engineer - Volume Rendering* 2010 - 2014

- Led architecture overhaul of volumetric capture and playback pipeline, reducing file sizes by 71% while maintaining sub-3mm geometric accuracy, enabling live-streamed photogrammetry for HoloLens enterprise applications.
- Established performance testing framework capturing 150+ metrics across 8 hardware platforms; identified and resolved critical memory leaks reducing system crashes by 87% in production deployments.
- Drove adoption of physically-based rendering for mixed reality environments across 5 AAA game studios; benchmarked solutions managing 2.8B polygons at 90FPS on mobile hardware.

Intel Visual Computing Institute — *Senior Research Engineer - 3D Reconstruction* 2006 - 2010

- Engineered real-time 3D mesh reconstruction algorithm processing multi-camera feeds at 60FPS; deployed in 15 broadcast production facilities for live virtual set composition.
- Published 6 technical papers on structure-from-motion optimization at CVPR and ICCV conferences; work cited 1,800+ times influencing commercial SLAM implementations industry-wide.
- Optimized photogrammetry processing pipeline reducing computation time from 8 hours to 22 minutes per asset, enabling production teams to increase scanning throughput by 340%.

EXTRACURRICULAR

IEEE Computer Society — *Executive Committee Member - Virtual Reality Technical Committee* 2019 - Present

- Chaired 3 international symposiums on immersive computing attracting 2,400+ researchers; curated 180+ peer-reviewed submissions establishing new conference standards.
- Established VR accessibility working group producing 12 technical standards adopted by W3C WebXR specification, benefiting 15M+ users with disabilities accessing XR content.

VRDC (Virtual Reality Developers Conference) — *Technical Track Chair & Keynote Speaker* 2015 - Present

- Delivered 8 keynotes and technical deep-dives reaching cumulative audience of 12,000+ developers; presentations ranked top 3% in attendee satisfaction scoring 4.8/5.0 average.
- Recruited and mentored 24 emerging speakers from underrepresented backgrounds in XR; 18 speakers advanced to invited conference presentations at top-tier venues.

TECHNICAL SKILLS

Languages – C++, C#, Python, GLSL, HLSL, JavaScript, Java

Technologies – SLAM, ARKit, ARCore, WebXR, OpenXR, Vulkan, Metal, CUDA, TensorFlow, PyTorch, photogrammetry, volumetric capture, spatial mapping, hand tracking, eye gaze estimation, depth estimation, neural rendering

Tools – Unity, Unreal Engine 5, Blender, OpenCV, PCL, RealSense SDK, Vuforia, Cesium.js, Git, Docker, Kubernetes, AWS, Azure Spatial Anchors